Tensor Decompositions for Deep Learning: Structures, Diagrams, and Parameter Counts

Notation and the dense baseline

Throughout, $\mathcal{D} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_N}$ is an N-way tensor—for instance a convolutional kernel or a weight matrix reshaped into higher order. Two aggregate quantities recur:

$$S := \sum_{i=1}^{N} n_i \qquad \text{(sum of mode sizes)}, \qquad\qquad \prod_{i=1}^{N} n_i \qquad \text{(dense storage — the baseline to beat)}.$$

Products used below

- **Outer product** $\mathbf{a} \circ \mathbf{b} \circ \mathbf{c}$: the rank-1 tensor with entries $a_i\, b_j\, c_k$.
- **Kronecker product** $\mathbf{A} \otimes \mathbf{B}$: every entry of $\mathbf{A}$ multiplies the whole block $\mathbf{B}$; sizes multiply.
- **Khatri–Rao product** $\mathbf{A} \odot \mathbf{B}$: the *column-wise* Kronecker product, $[\mathbf{a}_1 \otimes \mathbf{b}_1 \;\; \cdots \;\; \mathbf{a}_R \otimes \mathbf{b}_R]$; both matrices must have R columns.

In the diagrams below we use standard *tensor-network* notation: a node is a tensor, each line ("leg") is one index, a dangling leg is a free mode of size n_i, and a line joining two nodes is a summed (contracted) index whose size is the *rank* of that connection. The number of parameters of a decomposition is simply the total number of entries in all its nodes.

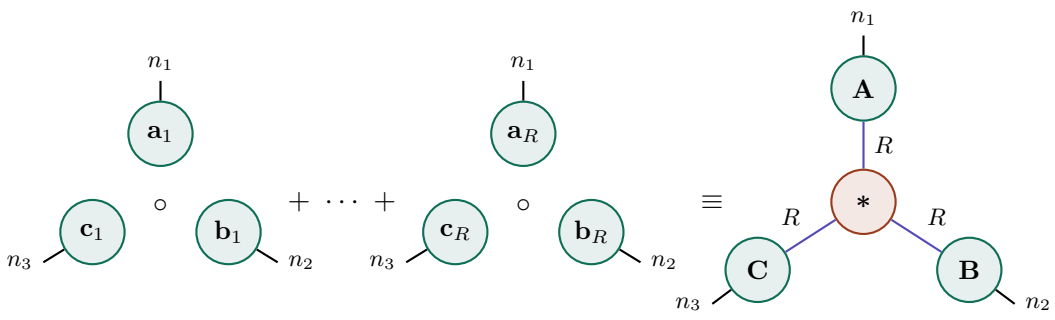
1 CP decomposition (rank R)

The tensor is a sum of R rank-1 terms,

$$\mathcal{D} \;=\; \sum_{r=1}^{R} \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r, \qquad d_{ijk} \;=\; \sum_{r=1}^{R} a_{ir}\, b_{jr}\, c_{kr},$$

and collecting the vectors as columns of factor matrices $\mathbf{A} \in \mathbb{R}^{n_1 \times R}$, $\mathbf{B} \in \mathbb{R}^{n_2 \times R}$, $\mathbf{C} \in \mathbb{R}^{n_3 \times R}$, the mode-1 unfolding takes the Khatri–Rao form

$$\mathbf{D}_{(1)} \;=\; \mathbf{A}\,(\mathbf{C} \odot \mathbf{B})^{\top}, \qquad \operatorname{vec}(\mathcal{D}) \;=\; (\mathbf{C} \odot \mathbf{B} \odot \mathbf{A})\,\mathbf{1}_R.$$



Left: sum of R rank-1 outer products. *Right:* equivalent network with a superdiagonal core $*$ ($\lambda_{rrr} = 1$) tying all factors to a *single shared* rank R.

Parameters. Each factor matrix costs $n_i R$ entries, so for general N

$$\#\text{params(CP)} \;=\; R\sum_{i=1}^{N} n_i \;=\; R\,S.$$

Linear in R and in S: the leanest structure in the table. The price: one global rank R must capture interactions across *all* modes at once, and the best rank-R approximation problem is ill-posed in general (degenerate solutions, swamps in ALS), so fitting is numerically finicky.

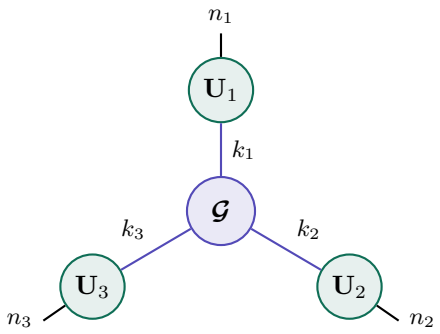
2 Tucker decomposition (multilinear ranks $k_1, \ldots, k_N$)

A small core $\boldsymbol{\mathcal{G}} \in \mathbb{R}^{k_1 \times \cdots \times k_N}$ is contracted with one factor matrix per mode, $\mathbf{U}_i \in \mathbb{R}^{n_i \times k_i}$:

$$\boldsymbol{\mathcal{D}} \;=\; \boldsymbol{\mathcal{G}} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \cdots \times_N \mathbf{U}_N, \qquad d_{ijk} \;=\; \sum_{p=1}^{k_1}\sum_{q=1}^{k_2}\sum_{s=1}^{k_3} g_{pqs}\, u^{(1)}_{ip} u^{(2)}_{jq} u^{(3)}_{ks},$$

which in vectorized form exposes the Kronecker structure of the table:

$$\text{vec}(\boldsymbol{\mathcal{D}}) \;=\; \big(\mathbf{U}_N \otimes \cdots \otimes \mathbf{U}_1\big)\ \text{vec}(\boldsymbol{\mathcal{G}}).$$



Identical topology to CP's network form, but the core $\boldsymbol{\mathcal{G}}$ is *dense* rather than superdiagonal, and each bond carries its *own* rank k_i.

Parameters.

$$\#\text{params(Tucker)} \;=\; \underbrace{\sum_{i=1}^{N} n_i k_i}_{\text{factors}} \;+\; \underbrace{\prod_{i=1}^{N} k_i}_{\text{core}}.$$

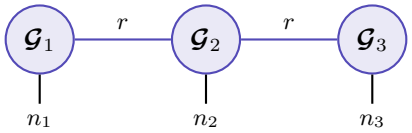
Per-mode ranks give fine-grained control, and the factors admit stable SVD-based fitting (HOSVD). But the core term $\prod_i k_i$ grows *exponentially* in the order N — the curse of dimensionality — so Tucker is best for low-order tensors ($N \leq 4$ or so).

3 Tensor Train (TT), rank r, shown for $N = 3$

Each entry is a product along an *open chain* of cores:

$$d_{ijk} \;=\; \mathbf{G}_1(i)\,\mathbf{G}_2(j)\,\mathbf{G}_3(k), \qquad \underbrace{\mathbf{G}_1(i)}_{1\times r}\ \underbrace{\mathbf{G}_2(j)}_{r\times r}\ \underbrace{\mathbf{G}_3(k)}_{r\times 1},$$

i.e. core $\boldsymbol{\mathcal{G}}_m$ has shape $r_{m-1} \times n_m \times r_m$ with boundary ranks $r_0 = r_N = 1$.



Parameters. Count entries core by core:

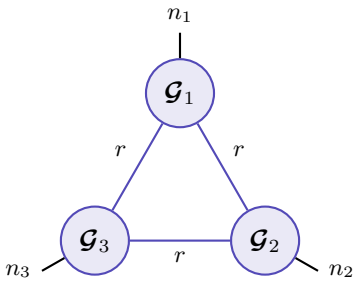
$$\underbrace{n_1 r}_{\mathcal{G}_1:\,1\times n_1\times r} \;+\; \underbrace{r\,n_2\,r}_{\mathcal{G}_2:\,r\times n_2\times r} \;+\; \underbrace{r\,n_3}_{\mathcal{G}_3:\,r\times n_3\times 1} \;=\; r\,(n_1+n_3)\;+\;r^2 n_2.$$

The asymmetry is the signature of the *open ends*: boundary cores are linear in r, interior cores quadratic. For general N the count is $\sum_m r_{m-1} n_m r_m$ — never a big joint core — so TT scales gracefully to high order, and TT-SVD gives stable quasi-optimal fitting.

4 Tensor Ring (TR), rank r

Close the chain into a ring; the boundary is removed by a trace:

$$d_{ijk} \;=\; \mathrm{Tr}\big(\mathbf{G}_1(i)\,\mathbf{G}_2(j)\,\mathbf{G}_3(k)\big), \qquad \text{every } \mathbf{G}_m(\cdot)\in\mathbb{R}^{r\times r}.$$



Parameters. All cores are now $r \times n_m \times r$, so every mode pays the interior price:

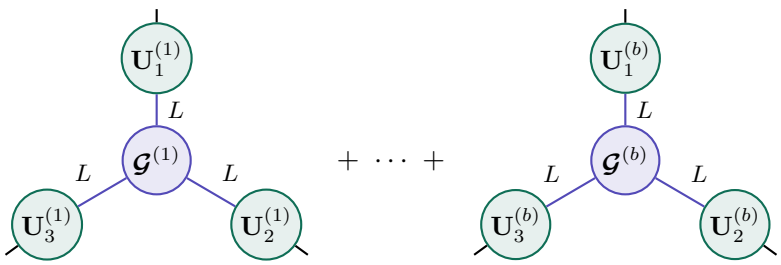
$$\#\text{params(TR)} \;=\; \sum_{m=1}^{N} r\,n_m\,r \;=\; r^2 \sum_{m=1}^{N} n_m \;=\; r^2 S.$$

Slightly more than TT at equal r, but all modes are treated symmetrically: the model is invariant to circular reordering of modes, and rank is distributed around the loop, which often buys more expressiveness per unit rank.

5 Block-Term Decomposition (BTD), b blocks of multilinear rank L

Instead of one big Tucker, sum b small Tucker terms with cores $\mathcal{G}^{(t)} \in \mathbb{R}^{L\times\cdots\times L}$:

$$\mathcal{D} \;=\; \sum_{t=1}^{b} \mathcal{G}^{(t)} \times_1 \mathbf{U}_1^{(t)} \times_2 \mathbf{U}_2^{(t)} \cdots \times_N \mathbf{U}_N^{(t)}.$$



Parameters. Each block costs a small Tucker: $L\sum_i n_i$ for the factors plus L^N for the core, hence

$$\#\text{params(BTD)} \;=\; b\left(L\,S \;+\; L^N\right).$$

BTD interpolates between the two extremes: with $L = 1$ each block is a rank-1 term and BTD reduces to CP with $R = b$; with $b = 1$ it *is* Tucker. It shines when the data genuinely contains several distinct low-multilinear-rank "sources."

6 Summary

Method	**Structure**	**Parameters**	**Scaling in order N**
Dense	all entries free	$\prod_i n_i$	exponential
CP (rank R)	$\sum_r \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r$ (Khatri–Rao)	$R\,S$	linear
Tucker	core $\times$ factors (Kronecker)	$\sum_i n_i k_i + \prod_i k_i$	exponential (core)
TT (rank r, N=3)	open chain of cores	$r(n_1{+}n_3) + r^2 n_2$	linear
TR (rank r)	closed ring of cores	$r^2 S$	linear
BTD (b blocks, rank L)	sum of small Tuckers	$b\,(LS + L^N)$	exponential (small cores)

Table 1: Parameter counts with $S = \sum_i n_i$. A leading multiplier K (e.g. number of layers or weight tensors being compressed) scales every row identically and is omitted.

Practical reading

CP is cheapest but hardest to fit (ill-posed approximation, one shared rank). **Tucker** is expressive with per-mode ranks but its core blows up with tensor order. **TT/TR** pay a quadratic factor r^2 in exchange for scalability to high N and stable SVD-based optimization; TR additionally buys mode symmetry. **BTD** buys multi-source structure at the cost of an extra hyperparameter, the block count b.